

What Can Each Study Claim?

Unit 1 · Topic 1.10 · THE PROBLEM

Suppose a student wants to know whether a new tutoring center raises test scores.

Plan A: Compare this year's test scores for students who choose to attend the center with scores for students who do not attend.

Plan B: Recruit 40 student volunteers, then use a coin flip to assign each volunteer either to attend the center or not to attend. Compare their later test scores.

Two plans for studying tutoring

Plan	Who?	Groups?	Response
A	Students	Choose	Scores
B	40 volunteers	Coin flip	Scores

FIRST TAKE — one sentence before you work the parts

Which plan gives stronger evidence that the center raises scores? Name the feature behind your choice.

- DOK 1** (a) Identify each plan as an observational study or an experiment.
- DOK 2** (b) Name one outside factor that could affect both choosing tutoring and test scores in Plan A. Explain both links in two sentences.
- DOK 3** (c) Which plan offers stronger evidence that tutoring causes a change in scores? Explain why its design helps with the outside factor from (b). Then state why its result may not apply to all students at the school.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.